Experiment 1

AIM: Explore GitHub/GitLab for open-source projects with different licenses

1. List of GitHub Licenses

	License Name	SPDX Identifier
1	MIT License	MIT
2	Apache License 2.0	Apache-2.0
3	GNU General Public License v3.0	GPL-3.0
4	GNU General Public License v2.0	GPL-2.0
5	GNU Lesser General Public License v2.1	LGPL-2.1
6	GNU Affero General Public License v3.0	AGPL-3.0
7	BSD 2-Clause (Simplified) License	BSD-2-Clause
8	BSD 3-Clause (New/Revised) License	BSD-3-Clause
9	Mozilla Public License 2.0	MPL-2.0
10	ISC License	ISC
11	Eclipse Public License 2.0	EPL-2.0
12	Creative Commons Zero v1.0 (Public Domain)	CC0-1.0
13	The Unlicense	Unlicense
14	Boost Software License 1.0	BSL-1.0
15	European Union Public License 1.2	EUPL-1.2

2. Licenses and Short Description Table

The table below provides a brief description of each major license to help developers choose the right one for their project

License	Short Description
MIT	Very permissive. Allows use, copy, modify, distribute, sell. Only requires keeping copyright notice.
Apache 2.0	Permissive. Like MIT but also grants patent rights and requires NOTICE file preservation.
GPL v3.0	Copyleft. Any modifications or distributions must remain open-source under GPL v3.
GPL v2.0	Older copyleft license. Similar to GPL v3 but without patent termination clauses.
LGPL v2.1	Weak copyleft. Libraries can be used in proprietary software without releasing source.
AGPL v3.0	Network copyleft. Even web-hosted software must release source code to users.
BSD 2-Clause	Very permissive. Allows reuse with attribution. No endorsement clause needed.
BSD 3-Clause	Like BSD 2-Clause but adds clause preventing use of project name for promotion.
MPL 2.0	Weak copyleft. Modified files must remain under MPL, but can combine with proprietary code.
ISC	Functionally equivalent to MIT/BSD. Very short and simple permissive license.
EPL 2.0	Weak copyleft used by Eclipse. Modifications to EPL code must be shared back.
CC0 1.0	Public domain dedication. Author waives all copyright globally.
Unlicense	Public domain equivalent. Author dedicates work to public domain without restrictions.
BSL 1.0	Permissive. Very simple; only requires keeping copyright notice. No advertising clause.
EUPL 1.2	European Union copyleft license, compatible with several other copyleft licenses.

3. Licenses Comparison Table

License	Commercial Use	Modify	Distribute	Patent Use	Private Use	Disclose Source
MIT	Legend	Legend	Legend	Permitted	Legend	Permitted
Apache 2.0	Legend	Legend	Legend	Legend	Legend	Permitted
GPL v3.0	Legend	Legend	Legend	Legend	Legend	Legend
GPL v2.0	Legend	Legend	Legend	Permitted	Legend	Legend
LGPL v2.1	Legend	Legend	Legend	Permitted	Legend	Not Permitted
AGPL v3.0	Legend	Legend	Legend	Legend	Legend	Legend
BSD 2/3	Legend	Legend	Legend	Permitted	Legend	Permitted
MPL 2.0	Legend	Legend	Legend	Legend	Legend	Not Permitted
ISC	Legend	Legend	Legend	Permitted	Legend	Permitted
CC0 1.0	Legend	Legend	Legend	Legend	Legend	Permitted
Unlicense	Legend	Legend	Legend	Permitted	Legend	Permitted

4. List of GitHub Alternatives

#	Platform	Type	Key Feature
1	GitLab	Open-source / SaaS	All-in-one DevOps platform with built-in CI/CD
2	Bitbucket	Proprietary (Atlassian)	Deep Jira & Trello integration
3	SourceForge	Free / Ad-supported	Long-standing host for legacy open-source projects
4	Gitea	Open-source (Self-hosted)	Lightweight, easy self-hosted Git service
5	Gogs	Open-source (Self-hosted)	Minimal footprint, runs on Raspberry Pi
6	Codeberg	Non-profit / Free	Community-owned, privacy-friendly, based on Gitea
7	Azure DevOps	Proprietary (Microsoft)	Enterprise integration with Azure cloud services
8	AWS CodeCommit	Proprietary (Amazon)	Managed Git service integrated with AWS ecosystem
9	Launchpad	Free (Canonical)	Ubuntu and Debian package hosting and collaboration
10	Phabricator	Open-source (Self-hosted)	Code review, task management, and wiki in one

5. GitHub vs GitLab Table

Feature	GitHub	GitLab
Founded	2008 (Owned by Microsoft)	2011 (Independent / Public)
License Model	Proprietary (Free + Paid tiers)	Open-core (Community Edition is open-source)
Self-hosting	GitHub Enterprise only (Paid)	Yes, GitLab CE/EE (Free Community Edition)
CI/CD	GitHub Actions	GitLab CI/CD (built-in, very mature)
Container Registry	GitHub Container Registry	GitLab Container Registry (built-in)
Issue Tracking	Basic Issues & Projects	Advanced Issues, Epics, Milestones, Boards
DevOps Pipeline	Partial (via integrations)	Full DevOps lifecycle in one platform
Security Scanning	Dependabot, Code Scanning	SAST, DAST, container scanning (built-in)
Pages (Static Sites)	GitHub Pages (free)	GitLab Pages (free)
Wiki	Per-repo Wiki	Per-repo and Group Wiki
Package Registry	GitHub Packages	GitLab Package Registry
Free Private Repos	Yes (unlimited)	Yes (unlimited)
Community / Ecosystem	Very large (most popular)	Growing, strong DevOps community
API	REST & GraphQL	REST & GraphQL
Pricing (Free tier)	Free for individuals & teams	Free tier with 400 CI/CD mins/month
Best For	Open-source, social coding	Enterprise DevOps, self-hosted teams

6. Open-Source vs Proprietary vs Freeware

Criteria	Open-Source	Proprietary	Freeware
Definition	Software with publicly available source code	Software owned by company/individual with restricted rights	Software free to use but source code is not available
Source Code	Publicly available, can be read & modified	Hidden / closed source	Not available to users
Cost	Usually free (some support/services cost)	Usually paid (subscription or one-time)	Free to use, no purchase required
License	OSI-approved licenses (MIT, GPL, etc.)	EULA – End User License Agreement	Custom freeware license
Modification	Allowed (per license terms)	Not allowed	Not allowed
Distribution	Allowed (per license terms)	Heavily restricted	Limited / not allowed
Commercial Use	Depends on license	Requires purchase/license	Usually for personal use only
Examples	Linux, Firefox, Python, VLC	Microsoft Windows, Adobe Photoshop, Oracle DB	Skype, CCleaner, Zoom (basic), 7-Zip
Support	Community / paid enterprise support	Official vendor support	Limited / community only
Security Audits	Anyone can audit the code	Only vendor performs audits	No public audit possible
Updates	Community or maintainer-driven	Vendor-controlled	Vendor-controlled, may be discontinued
Trust & Transparency	High – code is visible to all	Low – code is hidden	Medium – reputable vendors
Customization	Highly customizable	Not customizable	Not customizable